LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN
LMU MUNICH SCHOOL OF MANAGEMENT
INSTITUT FÜR RECHNUNGSWESEN UND WIRTSCHAFTSPRÜFUNG

Empirical Research in Accounting, Auditing and Capital Markets
Tutorial 2 · Data Preparation

Winter semester 2026/27 · M.Sc. / MBR
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Room 316 and Zoom, 29 October 2026

TUTORIAL 2 · THURSDAY, 29 OCTOBER 2026, 16:00–20:00

Data Preparation and Verification

I. Why data preparation matters
II. Identifiers
III. Data from an API: the SRN database
IV. Missing values and duplicates
V. Merges: rebuilding a Ball & Brown sample
VI. Auditing data and code
VII. Wrap-up

Today at a glance — mostly in RStudio

Time	Block	What you do	R Markdown
16:00	I · Why data preparation matters	Sample construction in Ball & Brown (1968) and Blankespoor et al. (2026)	—
16:20	II · Identifiers	Keys, names, ISIN checks	Part 1
16:35	III · Data from an API	SRN: token, request, pinned release, raw copy	Part 2
17:15	Break	10 minutes	
17:25	IV · Missing values and duplicates	Why is it missing? Is the key unique? Long → wide	Part 3
18:05	V · Merges	Link table, date conditions, anti_join, sample funnel	Part 4
19:00	Break	10 minutes	
19:10	VI · Auditing data and code	SRN vs. Compustat revenues; assertions; bug hunt	Parts 5–6
19:50	VII · Wrap-up	Checklist, outlook on Tutorial 3	Part 7

▶️ **IN RSTUDIO** **Pull the new material, copy the data from Moodle into data/, open session_02/Session_02_student.Rmd**

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Results are made in data preparation

Ball & Brown (1968)

- Three sources: **Compustat** (earnings), **CRSP** (returns), **Wall Street Journal** (announcement dates)
- Announcement dates “taken initially from the Wall Street Journal Index, then verified against the Wall Street Journal” (fn. 22)
- “Due to known errors in the data, not all firms could be included in all years” (fn. 23)
- Inclusion criteria leave **261 firms**

Blankespoor, Croom & Grant (2026)

- 412,192 investor questions to a GenAI chatbot
- Suggested questions identical except the ticker → ticker replaced by “TIC” to treat them as one
- Questions classified by a **machine-learning model** — about 75% agreement with human coders
- Removes incoherent, advice and app-support questions → **309,346** questions

Doing research yourself means: you decide — and document — every one of these steps.

Ball, R., & Brown, P. (1968). JAR 6(2), 159–178, pp. 167–168. Blankespoor, E., Croom, J., & Grant, S. M. (2026). JAE 82, 101908, Section 3.1.

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Every sample is a funnel — report it

Ball & Brown (1968): inclusion criteria

1. Earnings on Compustat for each year 1946–1966
2. Fiscal year ends December 31
3. Price data on CRSP for at least 100 months
4. Announcement dates in the Wall Street Journal

→ **261 firms**, fiscal years 1957–1965

Not included: young firms, failed firms, other year-ends

Blankespoor et al. (2026)	Questions
Random sample from the platform	412,192
− incomplete or incoherent	−22,678
− asking for explicit advice	−18,321
− app support / account questions	−61,847
Final sample (85,476 investors)	309,346

A funnel shows who is NOT in your sample. That is where bias hides.

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Identifiers: what makes a good key?

Identifier	Identifies	Source	Watch out
gvkey	firm	Compustat	6-digit text, leading zeros
permno	security	CRSP	one firm, several share classes
CUSIP	security	US/Canada	changes after name changes
ticker	security	exchanges	reused after delisting
ISIN	security	international	12 characters, country prefix
company name	—	everywhere	“BMW” ≠ “Bayerische Motoren Werke AG”

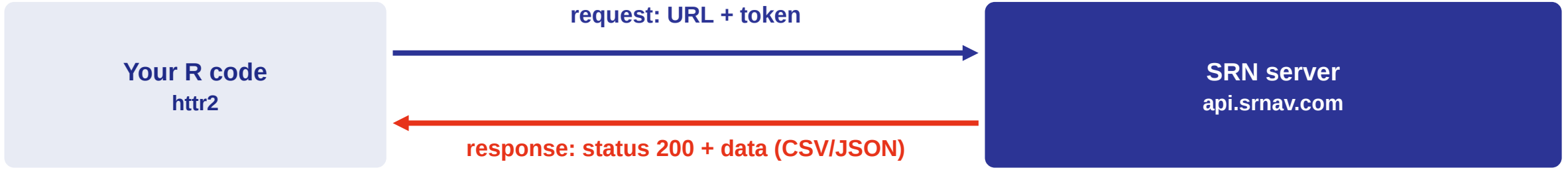
A key must be

- **unique** — one row per key value
- **stable** — does not change over time
- **clean** — same format in both data sets (text vs. number, spaces, case)

▶️ **IN RSTUDIO** — **Part 1 · Chunks 1.1–1.2: why names fail as keys; checking ISINs**

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What is an API? Your code asks, a server answers



Chunk 2.3

```
resp <- request("https://api.srnav.com") |>
  req_url_path_append("projects", "main-esrs-kpi",
                      "releases", "v2.4.csv") |>
  req_auth_bearer_token(Sys.getenv("SRN_TOKEN")) |>
  req_perform()
resp_status(resp)   # 200 = OK, 401 = token wrong
```

Three rules

- **Token** in `.Renviron`, never in your code or on GitHub
- **Pin the version** (`v2.4`, not `latest`)
- **Save the raw download** unchanged in `data/`

▶️ **IN RSTUDIO** — **Part 2 · Chunks 2.1–2.4: open endpoint, token, download, first look**

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Long and wide: the same data, two shapes

company	year	field	value
BMW	2024	revenues	142,380
BMW	2024	ghg_scope1	…
SAP	2024	revenues	34,176
SAP	2024	ghg_scope1	…

Long: one row per company × year × indicator



company	year	revenues	ghg_scope1
BMW	2024	142,380	…
SAP	2024	34,176	…

Wide: one row per company × year

Chunk 3.4

```
srn_wide <- srn_raw |> select(company_id, year, field_slug, value) |>
  pivot_wider(names_from = field_slug, values_from = value)
```

Illustrative values. Before pivoting: is company × year × field unique?

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Missing values: why is it missing?

1 Not reported

The firm chose not to disclose — itself a research question (voluntary disclosure).

2 Not applicable

The indicator does not apply — e.g. land use of a software firm.

3 Not collected

The database has not extracted it (yet) — a data problem, not a firm choice.

What you do with NAs changes the result

- `mean(x)` returns NA · `mean(x, na.rm = TRUE)` silently drops firms · replacing NA by 0 invents data
- First count them (`sum(is.na(x))`), then explain them, then decide — and write the decision down

▶️ **IN RSTUDIO** — **Part 3 · Chunks 3.1–3.2: count missing values; spot invisible problems in text (“Austria ”)**

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Duplicates: is the key really unique?

Chunk 3.3

```
cs |> count(gvkey, fyear) |> filter(n > 1)

# gvkey  fyear  n
# 001062  2010  2   <- INDL and FS format
# 001062  2011  2
```

Before you delete anything

- Look at one duplicate in full
- Find the variable that differs (`indfmt`)
- Decide by rule, not by `distinct()`

Typical reasons for duplicates

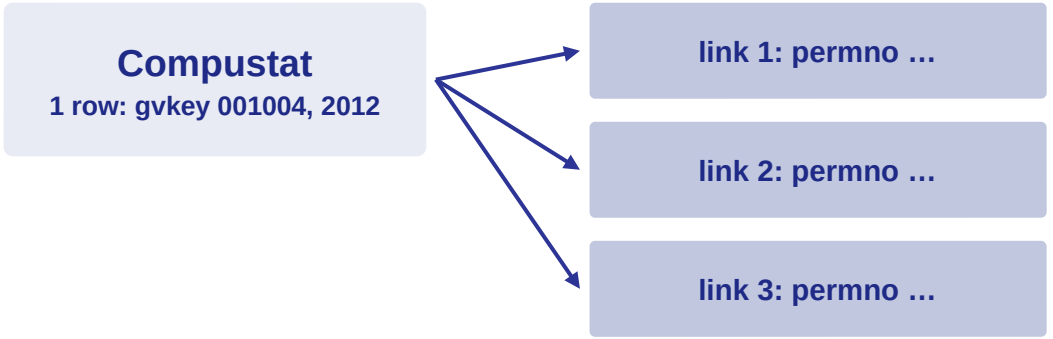
- Different formats (INDL/FS) · a change of fiscal year-end (two datadates in one fyear) · exact copies from the download (CRSP) · several securities per firm · the join itself (fan-out)

▶️ **IN RSTUDIO** — **Part 3 · Chunks 3.3–3.4 + Task A: unique keys, long → wide, how complete are the emission data?**

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Joins in five lines — and the fan-out trap

Function	Keeps
left_join(x, y)	all rows of x; NA if no match
inner_join(x, y)	only rows with a match
anti_join(x, y)	rows of x WITHOUT a match
full_join(x, y)	all rows of both



join by gvkey only → 3 rows for one firm-year

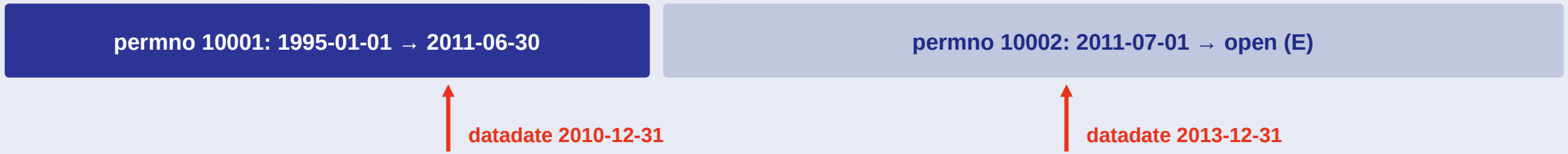
Golden rule: `nrow()` before and after every join · `anti_join()` to see who got lost · check the key again afterwards

A join that adds rows without you wanting it is the most common silent error in empirical research.

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Merging with dates: the CRSP–Compustat link table

gvkey 001234 over time



Chunk 4.3

```
bb_linked <- bb |>
  inner_join(link, by = join_by(gvkey,
             between(datadate, linkdt, linkenddt)))

bb |> anti_join(bb_linked, by = c("gvkey", "fyear"))
```

Standard filters

- `linktype` LU or LC (reliable links)
- `linkprim` P or C (primary security)
- open end date → treat as “today”

▶️ **IN RSTUDIO** — **Part 4 · Chunks 4.1–4.4: rebuild a Ball & Brown sample step by step**

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Our funnel: a Ball & Brown sample for 2010–2014

Step	Firms	Firm-years
Compustat INDL, fiscal year known	14,662	57,166
US firms reporting in USD	10,424	40,797
Earnings (ib) available	7,878	30,494
Fiscal year ends in December	5,980	22,921
Valid CRSP link at fiscal year-end	3,951	15,246
At least 60 monthly returns on CRSP	2,558	12,407

Who is lost?

- Non-US firms and other currencies
- Firms with other fiscal year-ends
- Firms without a CRSP link
- **Young and delisted firms** (short return histories) → survivorship bias

▶️ **IN RSTUDIO** — **Part 4 · Task B + bonus chunk 4.5: who gets lost — and do earnings news move returns?**

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Auditing generated data: SRN vs. Compustat revenues

Generated data are useful — and imperfect

- SRN quality snapshot: **74%** of reviewed values correct
- Blankespoor et al.: ML classification ≈ **75%** agreement with human coders
- Our check: revenues agree within 2% for **88% (AI 87%, human 90%)** of matched firm-years

Merge by ISIN and year

- SRN revenues (millions, reporting currency)
- Compustat Global `revt` (millions, `curcd`)
- Compare only same currency; ratio = SRN / Compustat

Reason for disagreement	How to spot it
Units (thousands vs. millions)	ratio ≈ 1,000 or 0.001
Currency	srn_currency ≠ curcd
Wrong year	ratio ≈ 1 for the neighbouring year
Definition / extraction error	look it up in the report — who is right?

▶️ **IN RSTUDIO** — **Part 5 · Chunks 5.1–5.3 + Task C: merge by ISIN, compare, look up one disagreement**

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Assertions and the bug hunt

An assertion stops your code when something is wrong

```
# key unique?
stopifnot(!any(duplicated(d[c("gvkey", "fyear")])))
# no rows added by the join?
stopifnot(nrow(after) == nrow(before))
# identifier in the right format?
stopifnot(all(nchar(d$gvkey) == 6))
```

Bug hunt: an AI-written merge script

- Round 1 (10′): by hand, mark every bug with `# BUG:`
- Round 2 (10′): ask an AI tool — code only, never the data
- Round 3 (5′): which bugs did the AI miss — or invent?

What the AI cannot know unless you tell it

- That gvkey has leading zeros · that links are valid only for certain dates · that SRN revenues are in millions · that names are not unique keys

▶️ **IN RSTUDIO** — **Part 6 · Chunks 6.1–6.2: assertions, bug hunt and corrected version**

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What you can do now — and what comes next

- ☐ I can name the key of a data set and check that it is unique
- ☐ I can download data from an API — with a token that never appears in my code
- ☐ I ask why a value is missing before I decide what to do with it
- ☐ I compare row counts before and after every join and use anti_join to see who got lost
- ☐ I can build a sample funnel and name the firms that drop out
- ☐ I check generated data and generated code against an independent source

Before 2 Nov

Paper Brief on Ball & Brown (1968)

Before 5 Nov

Commit and push your completed Tutorial 2

Thu 5 Nov

Tutorial 3 · Measurement and construct validity

By Sat 7 Nov

Firm commitment by e-mail and ISC registration

Today’s result: a merged, checked data set — and a funnel that says who is missing.

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QUESTIONS?

See you on 5 November

Bring your merged data set from today.

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